

DATA VISUALIZATION USING POWER BI

AIM

To visualize and analyze the E-Commerce dataset using Line Chart, Bar Chart, Pie Chart, Histogram, Scatter Plot, and Box Plot with Matplotlib and Seaborn to understand price trends, country-wise comparisons, product category distribution, price patterns, and relationships between e-commerce attributes.

ALGORITHM

Step 1: Input Data Loading

- Read the E-Commerce dataset containing columns such as product_name, country, category, unit_price, quantity, order_date, and subscription_status.
- Load the cleaned E-Commerce CSV file into Power BI.

Step 2: Country Filter Check

- Check slicer selection for Country.
- Default selected countries: all available countries.
- Retain all rows matching the selected countries in Filtered_Data.

Step 3: KPI Total Quantity Calculation

- Sum all values in the quantity column from Filtered_Data.
- Total Quantity = SUM(quantity)
- Display the result as the Total Quantity KPI card.

Step 4: Bar Chart Calculation — Average Unit Price by Country

- Group Filtered_Data by country.
- Calculate the average unit_price for each country.
- Sort countries according to their average unit price.
- Display the results using a bar chart.

Step 5: Pie Chart Calculation — Quantity by Category

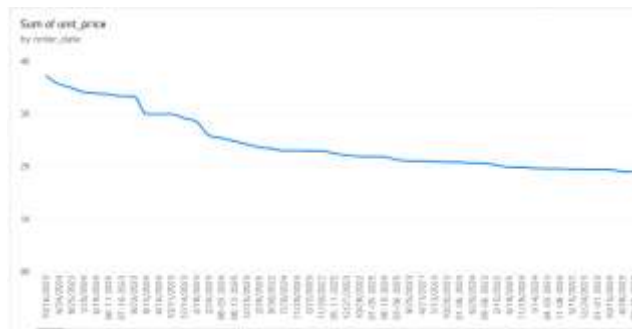
- Group Filtered_Data by category.
- Calculate the total quantity for each category.
- Calculate the percentage share of each category relative to total quantity.
- Display the category-wise distribution using a pie chart.

Step 6: Line Chart Calculation — Unit Price Trend

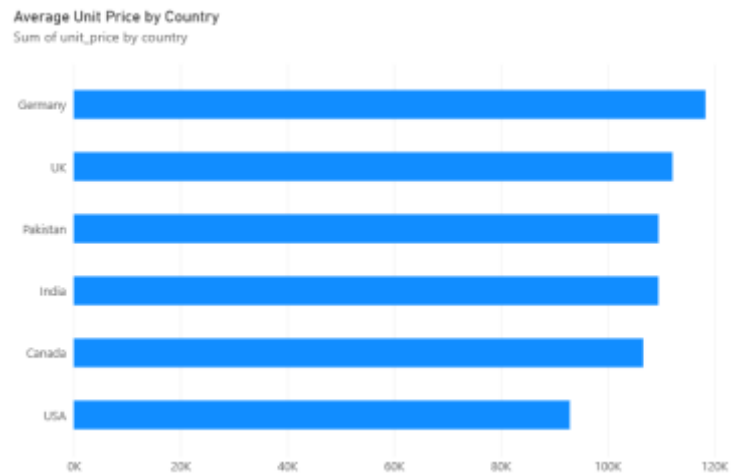
- Group Filtered_Data by order_date.
- Calculate the average unit_price for each date.
- Plot the values chronologically using a line chart.
- Display the trend of average unit price over time

OUTPUT

LINE CHART

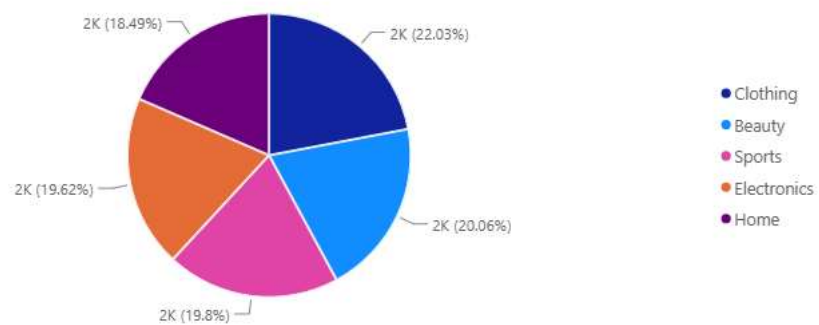


BAR CHART



PIE CHART

Sum of quantity
by category



RESULT :